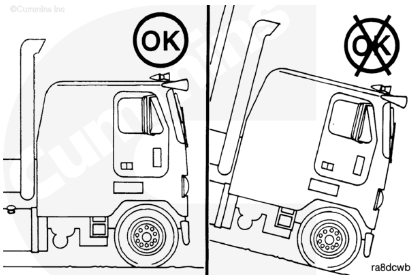
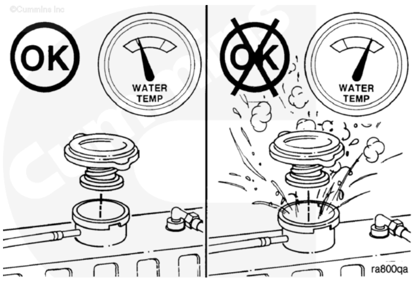
Trainings

Drain

Position the equipment on level ground.

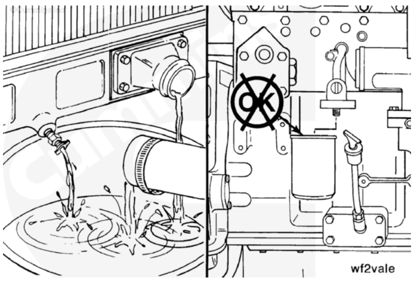




⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

Remove the pressure cap after the engine is cool.



⚠ WARNING ⚠

Coolant is toxic. Keep away from children and animals. If not reused, dispose of in accordance with local environmental regulations.

Do **not** allow the cooling system to dry out.

If the coolant is to be reused, the container **must** be free of oil and dirt. Before pouring coolant back into the engine, it **must** be tested for SCA concentration. Refer to Procedure 018-018 in Section V. (/qs3/pubsys2/xml/en/procedures/269/269-018-018.html)

Open the radiator draincocks.

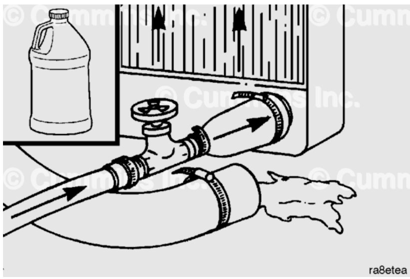
Remove the lower radiator hose(s).

Do **not** remove the coolant filter.

Flush

⚠ CAUTION ⚠

Do not use caustic cleaners in the cooling system. Aluminum components will be damaged.

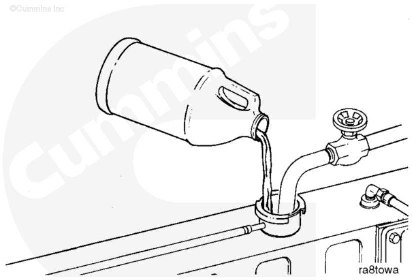


The cooling system **must** be clean to work correctly and to eliminate buildup of harmful chemicals.

Restore™ is a heavy-duty cooling system cleaner that removes corrosion, silica gel, and other deposits. The performance of Restore™ is dependent on time, temperature, and concentration levels. An extremely scaled or flow-restricted system, for example, can require higher concentrations of cleaners, higher temperatures, or longer cleaning times, or the use of Restore Plus™. Up to twice the recommended concentration levels of Restore™ can be used safely. Restore Plus™ **must** be used **only** at its recommended concentration level. Extremely scaled or fouled systems can require more than one cleaning.

⚠ CAUTION ⚠

Fleetguard® Restore™ contains no antifreeze. Do not allow the cooling system to freeze during the cleaning operation.

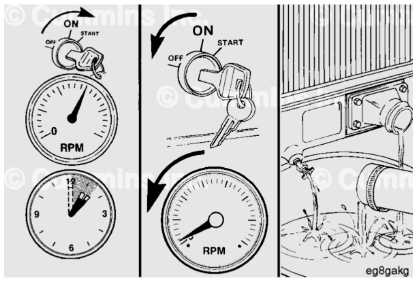


After draining the coolant, immediately add 3.8 liters [1 gal] of Fleetguard® Restore™, Restore Plus™, or equivalent, for each 38 to 57 liters [10 to 15 gal] of cooling system capacity, and fill the system with plain water.

Turn the cab heater temperature switch to high to allow maximum coolant flow through the heater core. The blower does **not** have to be on.

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



⚠ WARNING ⚠

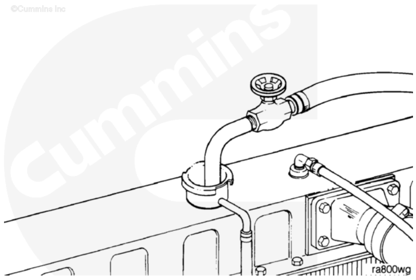
Coolant is toxic. Keep away from children and animals. If not reused, dispose of in accordance with local environmental regulations.

Operate the engine at normal operating temperatures, at least 85°C [185°F], for 1 to 1½ hours.

Shut the engine off and allow it to cool to 50°C [120°F].

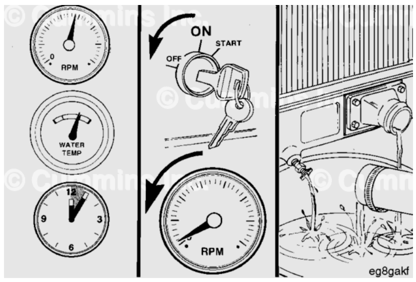
Drain the cooling system.

Fill the cooling system with clean water.



⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.



Operate the engine at high idle for five minutes with the coolant temperature above 85°C [185°F].

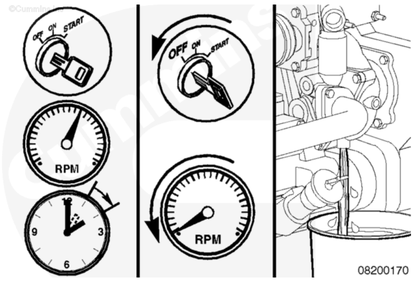
Shut the engine OFF, allow to cool to 50° C [120°F], and drain the cooling system.

If the water being drained is still dirty, the system **must** be continually flushed until the water is clean.

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam

can cause personal injury.



⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

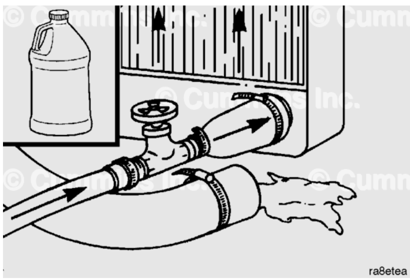
Operate the engine at high idle for 5 minutes with the coolant temperature above 85°C [185°F].

Shut the engine off, allow to cool to 50°C [120°F] and drain the cooling system.

If the water being drained is still dirty, the system **must** be continually flushed until the water is clean.

⚠ CAUTION ⚠

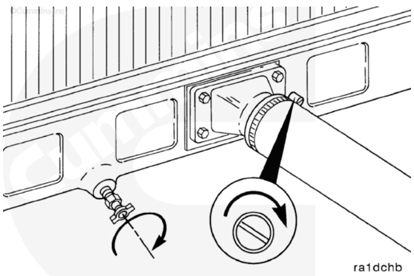
To reduce the possibility of engine damage, do not add coolant while the engine is hot



Fill the cooling system with fully-formulated coolant or a 50/50 mixture of fully-formulated antifreeze and good-quality water. Use a service filter to bring the coolant to the correct SCA concentration level. Refer to Procedure 018-018 in Section V. (/qs3/pubsys2/xml/en/procedures/269/269-018-018.html)

Fill

Close the radiator draincocks.



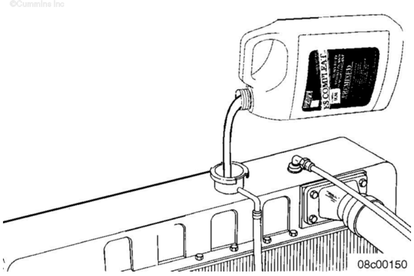
Install the lower radiator hose(s).

Tighten the hose clamps.

Torque Value: 5 n•m [44 in-lb]

⚠ CAUTION ⚠

Do not add coolant to a hot engine. Engine castings can be damaged. Allow the engine to cool below 50°C [120°F] before adding coolant.



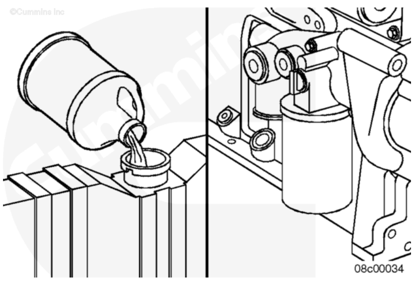
Cummins Inc. recommends using either a 50/50 mixture of good-quality water and fully-formulated antifreeze, or fully-formulated coolant when filling the cooling system. The fully-formulated antifreeze or coolant **must** meet TMC RP329 or TMC RP330 specifications.

Good-quality water is important for cooling system performance. Excessive levels of calcium and magnesium contribute to scaling problems, and excessive levels of chlorides and sulfates cause cooling system corrosion.

Water Quality	
Calcium Magnesium (Hardness)	Maximum 170 ppm as ($CaCO_3$ + $MgCO_3$)
Chloride	40 ppm as (Cl)
Sulfate	100 ppm as (SO_4)

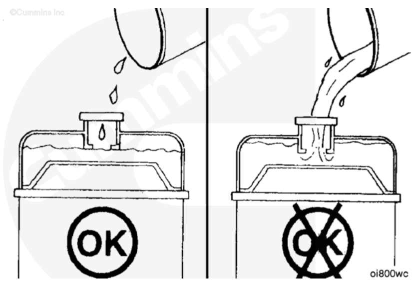
Cummins Inc. recommends using Fleetguard® COMPLEAT ES™. It is available in glycol forms (ethylene and propylene) and complies with TMC RP329 and RP330 standards.

Fill the cooling system with heavy-duty coolant and install the correct service filter. Refer to Procedure 018-018 in Section V. (/qs3/pubsys2/xml/en/procedures/269/269-018-018.html)



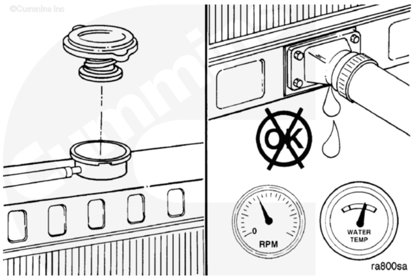
Note : Some radiators have two fill locations, both of which must be filled when the cooling system is drained.

Fill the cooling system with coolant to the bottom of the fill neck in the radiator fill (or expansion) tank.

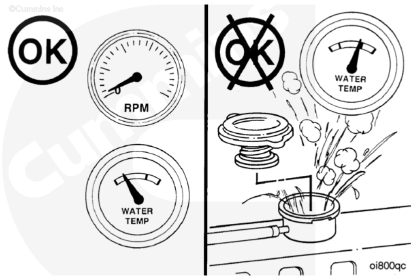


Install the radiator or expansion tank fill cap.

Operate the engine until it reaches a temperature of 82°C [180°F], and check for leaks.



Shut the engine OFF and allow it to cool.



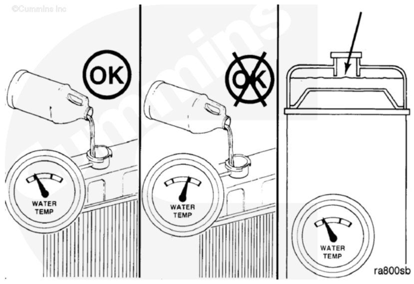
⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ CAUTION ⚠

Do not add cold coolant to a hot engine. This can cause engine casting damage. Allow the engine to cool to below 50°C [120°F] before adding coolant.

Remove the radiator or fill cap and check the coolant level. Refer to the original equipment manufacturer (OEM) service manual for recommendations. Add coolant if necessary. Do **not** overfill.

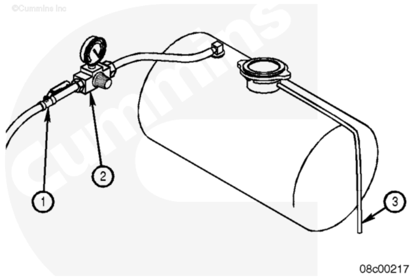


Pressure Test

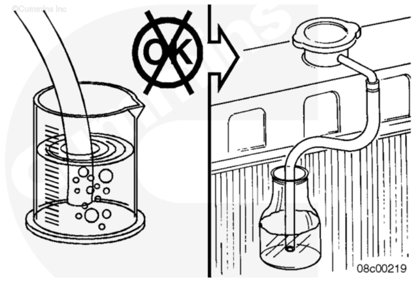
Connect the air line in place of a vent line on the surge tank.

Install an air regulator (2) between the ball valve (1) of the air line and the surge tank connection.

Attach a hose to the radiator overflow connection (3).



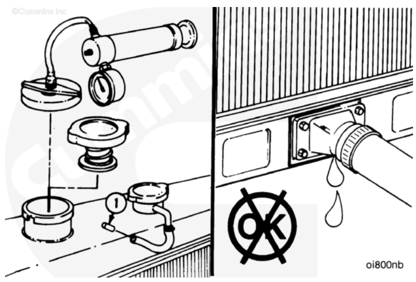
Place the free end of the hose below the water level in a container of water.



⚠ CAUTION ⚠

Do not apply more than 140 kPa [20 psi] air pressure to the cooling system. The water pump seal can be damaged.

Pressurize the cooling system slowly to the rated radiator cap pressure until bubbles can be seen in the overflow bottle.



If the bubbles start at a lower pressure than the minimum recommended cap pressure (See cooling system specifications), inspect the radiator pressure cap. Refer to Procedure 008-047 in Section 8. (/qs3/pubsys2/xml/en/procedures/269/269-008-047-tr.html)

Replace the cap, if necessary.